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Two simple differential equations

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1 The one-dimensional Poisson equation

The first equation we looked at is

$$\frac{d^2u}{dx^2} = f(x) \qquad 0 < x < L \tag{1}$$

with boundary conditions

$$u(0) = a\,, \qquad u(L) = b\,. \tag{2}$$

This is a very elementary example in one space dimension of the so-called Poisson equation. To solve it we integrate once to find

$$\frac{du}{dx} = A + \int_0^x f(y)dy \tag{3}$$

where A is the unknown value of du/dx at $x = 0$; we will determine it later. Let us set temporarily

$$F(x) = \int_0^x f(y)dy \tag{4}$$

so that

$$\frac{du}{dx} = A + F(x)\,. \tag{5}$$

Now we integrate again

$$u = Ax + \int_0^x F(y)dy + a \tag{6}$$

where we have imposed the known condition $u(x = 0) = a$. Now we note that, by integrating by parts,

$$\begin{aligned}\int_0^x F(y)dy &= [yF(y)]_{y=0}^{y=x} - \int_0^x y\frac{dF}{dy}dy \\ &= xF(x) - \int_0^x yf(y)dy = x\int_0^x f(y)dy - \int_0^x yf(y)dy \\ &= \int_0^x (x-y)f(y)dy \end{aligned} \tag{7}$$

Thus we have found that

$$u = a + Ax + \int_0^x (x-y)f(y)dy \tag{8}$$

In order to find A we need to impose the known condition at $x = L$:

$$b = a + AL + \int_0^L (L-y)f(y)dy \tag{9}$$

from which

$$A = \frac{1}{L}\left[b - a - \int_0^L (L-y)f(y)dy\right]\,. \tag{10}$$

1. Example 1: Take $f(x) = x^n$. Then

$$\int_0^x (x-y) f(y) dy \;=\; \int_0^x (x-y) y^n dy \;=\; x\int_0^x y^n dy - \int 0^x y^{n+1} dy \;=\; x\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2}$$
$$=\; x^{n+2}\left(\frac{1}{n+1} - \frac{1}{n+2}\right) \;=\; \frac{x^{n+2}}{(n+1)(n+2)} \,. \tag{11}$$

In this case therefore

$$A \;=\; \frac{1}{L}\left(b - a - \frac{L^{n+2}}{(n+1)(n+2)}\right) \,, \tag{12}$$

and the final solution is

$$u \;=\; a + \left(b - a - \frac{L^{n+2}}{(n+1)(n+2)}\right)\frac{x}{L} + \frac{x^{n+2}}{(n+1)(n+2)} \;=\; a + (b-a)\frac{x}{L} + \frac{x}{(n+1)(n+2)}\left(x^{n+1} - L^{n+1}\right) \,. \tag{13}$$

2. Example 2: Take $f(x) = \cos qx$. Then

$$\int_0^x (x-y) f(y) dy \;=\; \int_0^x (x-y)\cos qy dy \;=\; x\int_0^x \cos qy dy - \int_0^x y\cos qy dy$$
$$=\; x\frac{\sin qx}{q} - \frac{\cos qx - 1}{q^2} + \frac{x}{q}\sin qx \;=\; \frac{2x}{q}\sin qx - \frac{\cos qx - 1}{q^2} \tag{14}$$

In this case

$$A \;=\; \frac{1}{L}\left[b - a - \frac{2L}{q}\sin qL + \frac{\cos qL - 1}{q^2}\right] \,, \tag{15}$$

etc.

2 The one-dimensional Helmholtz equation

The simplest Helmholtz equation in one dimension is

$$\frac{d^2 u}{dx^2} - \lambda u \;=\; f(x) \qquad 0 < x < L \tag{16}$$

with the same general boundary conditions as before

$$u(0) \;=\; a \,, \qquad u(L) \;=\; b \,. \tag{17}$$

Let us consider the *homogeneous* case, in which $f = 0$. Suppose that $\lambda = \alpha^2 > 0$. We note that

$$\frac{d}{dx}e^{\pm\alpha x} \;=\; \pm\alpha e^{\pm\alpha x} \,, \qquad \frac{d^2}{dx^2}e^{\pm\alpha x} \;=\; (\pm\alpha)^2 e^{\pm\alpha x} \;=\; \alpha^2 e^{\pm\alpha x} \tag{18}$$

Thus we find that $e^{\alpha x}$ and $e^{-\alpha x}$ are both solutions of the equation. These solutions do not satisfy the boundary conditions, but we can assemble a solution that does satisfy the boundary conditions by combining them:

$$u \;=\; A_1 e^{\alpha x} + A_2 e^{-\alpha x} \tag{19}$$

and determine A_1 and A_2 in such a way that the boundary conditions are satisfied:

$$a \;=\; A_1 + A_2 \,, \qquad b \;=\; A_1 e^{\alpha L} + A_2 e^{-\alpha L} \tag{20}$$

For example, if $a = 0$, we find

$$A_2 \;=\; -A_1\,, \qquad A_1\left(e^{\alpha L} - e^{-\alpha L}\right) \;=\; b\,, \tag{21}$$

from which

$$A_1 \;=\; \frac{b}{e^{\alpha L} - e^{-\alpha L}} \tag{22}$$

or, recalling that

$$\sinh \alpha L \;=\; \frac{1}{2}\left(e^{\alpha L} - e^{-\alpha L}\right) \tag{23}$$

$$A_1 \;=\; \frac{2b}{\sinh \alpha L} \tag{24}$$

Substituting into (19) we then have

$$u \;=\; \frac{2b}{\sinh \alpha L} e^{\alpha x} - \frac{2b}{\sinh \alpha L} e^{-\alpha x} \;=\; b\frac{\sinh \alpha x}{\sinh \alpha L}\,. \tag{25}$$

If in (16) $\lambda = -\beta^2$ so that

$$\frac{d^2u}{dx^2} + \beta^2 u \;=\; 0 \qquad 0 < x < L \tag{26}$$

we find

$$u \;=\; A_1 \cos \beta x + A_2 \sin \beta x \tag{27}$$

and determine A_1 and A_2 as before

$$a \;=\; A_1\,, \qquad b \;=\; A_1 \cos \beta L + A_2 \sin \beta L \tag{28}$$

from which

$$A_1 \;=\; a\,, \qquad A_2 \;=\; \frac{b - a\cos \beta L}{\sin \beta L} \tag{29}$$